Sophia Shantharupan

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Education

McMaster University Hamilton, ON
Bachelor of Applied Science - Honours Computer Science *Sept 2022 - Apr 2026*

- Courses: Data Structures & Algorithms, Databases, Algorithms & Software Design, Algorithms & Complexity
- McMaster University Dean's Honour List 2023 - 2025

Experience

Software Developer | *TypeScript, React, Python, Flask-SocketIO, Phaser3* Sept 2025 - Present
McMaster University Baby Lab *Remote*

- Engineered a real-time multiplayer networking system **achieving <7ms average latency for player position updates**, by designing an O(1) hash map–based player-to-room lookup structure, and implementing bi-directional Flask-SocketIO communication.
- **Improved opponent position desynchronization by up to 99%** by accounting for player deceleration mechanics to provide real-time positions of all players in a game room.
- **Eliminated 100% of identified invalid input errors** and reduced client-side input vulnerabilities by developing and enforcing a robust input validation algorithm.

Research Assistant II | *SQL, PowerBI, Dax, Power Automate* Oct 2023 - Present
McMaster University Advancement Services *Remote*

- **Achieved a 90% reduction in prospect lookup time** by engineering an MS SQL database and Power BI CRM dashboard with DAX, built from unstructured text data.
- Composed Power Automate flows to facilitate automatic dashboard updates when SharePoint Excel data is edited.
- **Decreased manual data verification by 95%** for selecting a prime alumni event hosting location, by engineering a CRM dashboard using Power BI.

Website Developer | *JavaScript, CSS, HTML* Feb 2022 - June 2022
BeeOneSmartie - Online Tutoring Company *Vaughan, ON*

- Applied Agile methodologies with **JavaScript**, **HTML** and **CSS** in a five-member team, enhancing the site through 4+ sprints to meet client specifications.
- **Improved accuracy of user's final score by 33%** by utilizing an HTML tag to clear the user's input each time an answer is submitted.
- Resolved a critical error within the project's JavaScript algorithm to allow a new, randomly generated math question to be presented to the user each time the correct answer is submitted.

Projects

Language Learning Mobile Game | *C#, Unity, Python, AWS (DynamoDB, S3 and Lambda)*

- Refactored C# question randomization algorithm, eliminating constraint on minimum question set size, and reducing runtime by up to 99%.
- Reduced data entry and processing time by ~95% by creating a Lambda function to automate new DynamoDB table entries upon batch file upload to an S3 bucket.
- Achieved 99% reduction of manual data entry for the creation of all datasets, using Python scripting.

</> **Unix Shell and History Feature** | *C, Ubuntu*

- C program developed on Ubuntu to model a shell. Supports 100% of the system calls provided by fork() and exec().
- Prevented 100% of discovered invalid command crashes with custom error handling for invalid inputs.
- Optimized execution by designing the !! operator to re-execute the latest command, decreasing redundant input, and reducing command re-entry time by ~85%.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C, C#, SQL, HTML, CSS
Tools & Frameworks: Git, Unity, MonogoDB, Postman, Power BI, Power Automate, JUnit, MS SQL Server
Libraries: React, Node.js, Pandas, NumPy, Matplotlib, Sklearn
Other Courses: Operating Systems, Differential Privacy, Linear Optimization, Computer Networks & Security